

Post AI Era Datacenters Deepdive

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AI Boom

What's the driving force of datacenter construction and expansions?

Artificial Intelligence

- Datacenters aren't new
- Many ambitious datacenter projects
- Push towards AGI
- Technology replacing humans in most fields



Resulting Race 🏁

- Companies are making huge investments
- Scale their Large Language model
- Win the race to AGI
- Obtain the largest possible market share

Energy Requirements ⚡

Ways to fulfil the insatiable hunger of datacenters for electricity

Distributed Generation

City main line isn't enough. This necessitates the use of local energy generation to meet energy demands.

Renewable energy

- Solar panels
- Wind turbines
- Geothermal
- Nuclear

Fossil fuels (56%)

- Coal (mainly China) ⁵³
- Diesel
- Gas
 - Field gas
 - CNG
 - LPG
 - Low BTU gases

Do Datacenters fit the definition?

Distributed Generation $\equiv$ On-site Generation

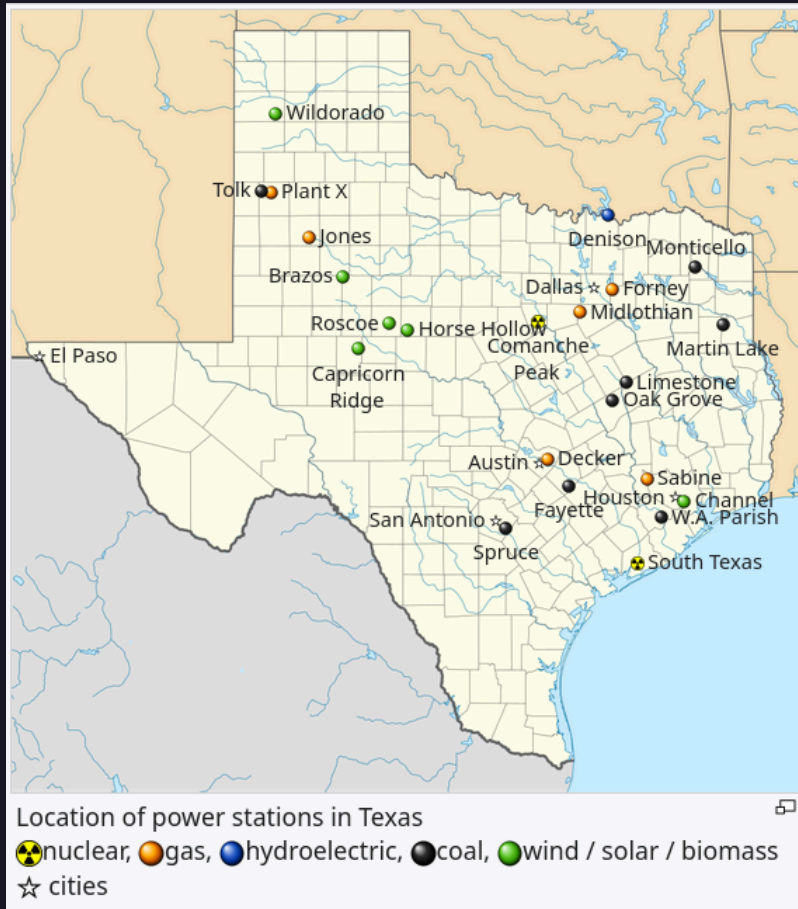
- Technologies being used can be considered D.E.R. (Distributed Energy Resources)
E.g. Solar, Wind, mini gas turbines etc.

But!

- The sheer scale of them though doesn't necessarily fit with the definition "**small**, grid-connected or distribution system-connected devices" ⁴⁹



Scale close or exceeding centralized



Examples

Descending Capacity Order

Name	Capacity(MW)
Sabine River Works	2,124.0
Forney Energy Center	1,784.0
Midlothian	1,734.0
Temple 1&2	1,500.0

Fossil fuel candidates

Turbines can be powered by:

- **Diesel**
- **Field gas**: Gas extracted from a production well before it enters a natural gas processing plant ³¹.
- **CNG**: Compressed Natural Gas, mainly methane compressed at a pressure of 200 to 248 bars ³².
- **LPG**: Liquefied Petroleum Gas, a mixture of propane and butane liquefied at 15 °C and a pressure of 1.7-7.5 bars ³².
- **LowBTU gases**: Natural gas, which at the wellhead has a gross heating value ≤ 450 BTU. Part of petroleum refining and crude oil, natural gas production.

Fuel Comparison

Comparison	CNG	LPG
Constituents	Methane	Propane, Butane
Energy density	Lower caloric value	Higher caloric value
Cost	Cheaper	More expensive
Emissions	Less than LPG	Less than gasoline
Safety	Disperses quickly	Settles to ground, but highly inflammable

CNG risk of ignition is low so even though LPG is highly inflammable it is the more likely of the two to ignite.

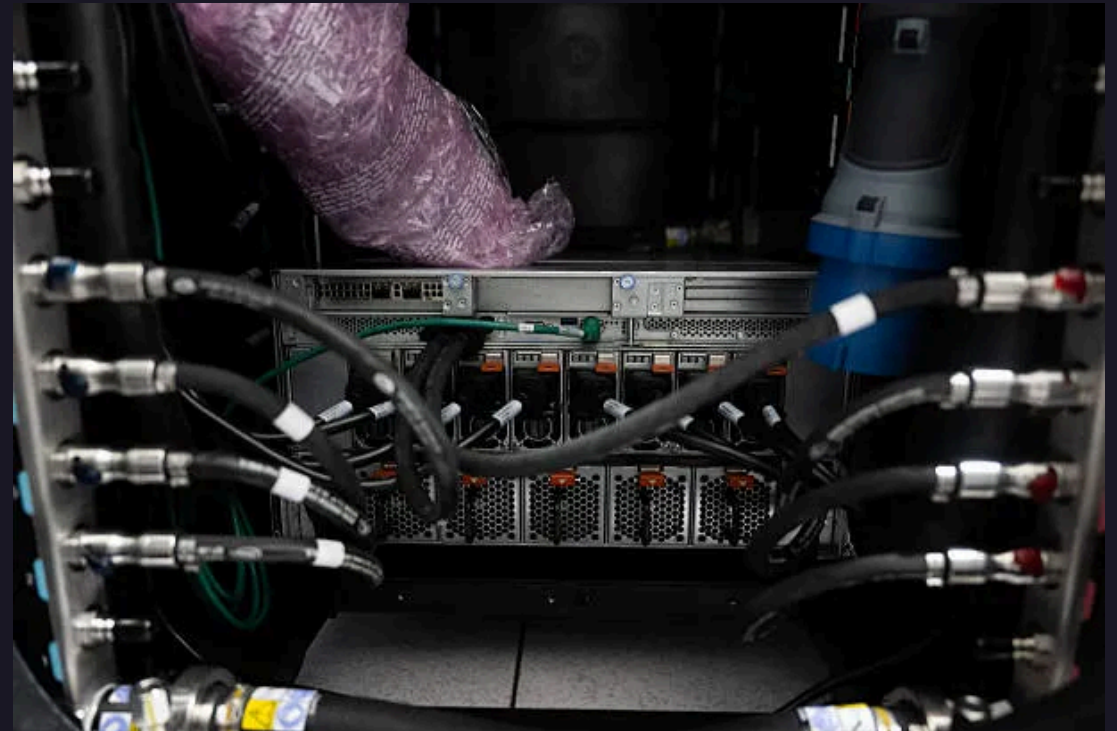
Cooling ❄️

Water 💧 vs Air 🌀

Water is chosen because it is more thermally conductive than air

- Air: 0.026 [W/mK]
- Water: 0.6089 [W/mK] ³⁷

Thus the more energy efficient solution depend on a more and more direct heat dissipation from water or even special non-conductive liquids.



Thermal conductivity δ°

The thermal conductivity of a material is a measure of its ability to conduct heat. It quantifies the proportionality between the heat flux q and the temperature gradient ∇T in the direction of heat transport.

$$q = -k\nabla T \text{ [W/mK]}$$

- $q \text{ [W/m}^2\text{]}$: heat flow rate per unit area
- $\nabla T \text{ [K/m]}$: temperature gradient in the direction of heat transport



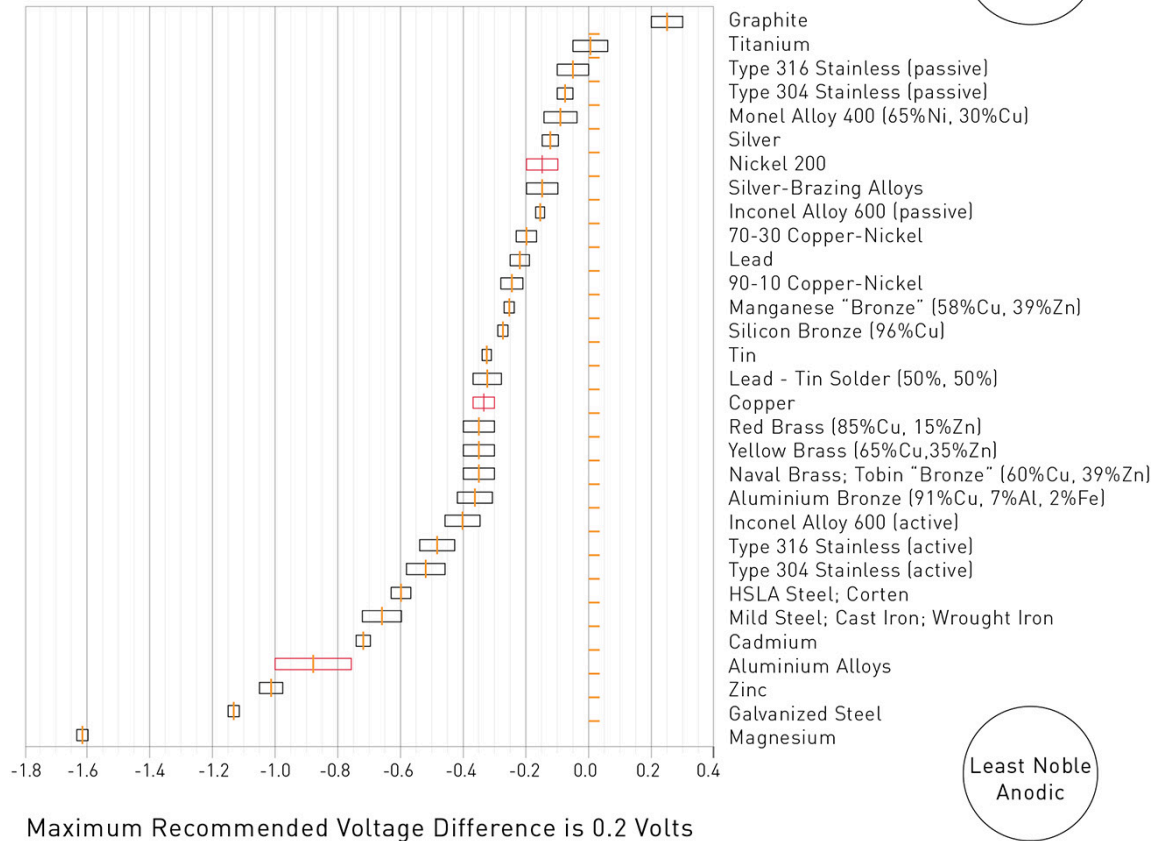


Copper vs Aluminum

- Copper piping
-  **Great thermal conductivity**³³
 - Copper: $401 [W/mK]$
 - Aluminum: $205 [W/mK]$
-  **More expensive per tonne**³⁴
 - Copper: $12808\$/t$
 - Aluminum: $3385\$/t$

The Galvanic Series Chart

Average Voltage in Seawater



Galvanic Corrosion ⚠

Galvanic is an electrochemical process that occurs when two metals with different electrochemical activity are in contact with each other (like copper and aluminum).



Galvanic Corrosion ⚠

The more noble, passive metal (copper or nickel) drives the corrosion of the active, less noble metal (aluminum), where the passive metal remains fairly unharmed. ³³

Cooling types

1. Computer Room Air Conditioning

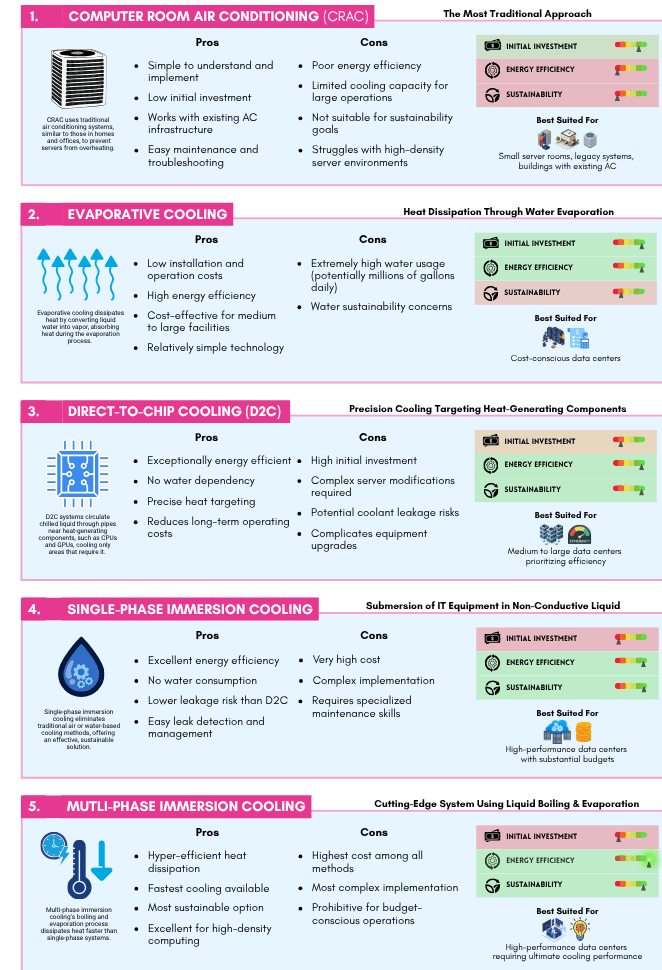
2. Evaporative Cooling

3. Direct-To-Chip Cooling

4. Single-Phase Immersion Cooling

5. Multi-Phase Immersion Cooling ³⁸

Data Center Cooling Methods Compared



Computer Room Air Conditioning (CRAC)

The origin of datacenter cooling. Rooms dedicated to housing IT equipped with traditional home use A.C. units.

Metrics	Rating
Energy Efficiency ⚡	1/5 ■
Sustainability 🌱	1/5 ■
Initial Investment 💰	4/5 ■



13KW CRAC

Evaporative Cooling

Dissipates heat by converting liquid water into vapor, absorbing heat during the evaporation process.

2 ways:

- **Direct** evaporative cooling, where server room air passes through a water-soaked membrane
- **Indirect** evaporative cooling, which uses heat exchangers to cool air before pumping it into server rooms

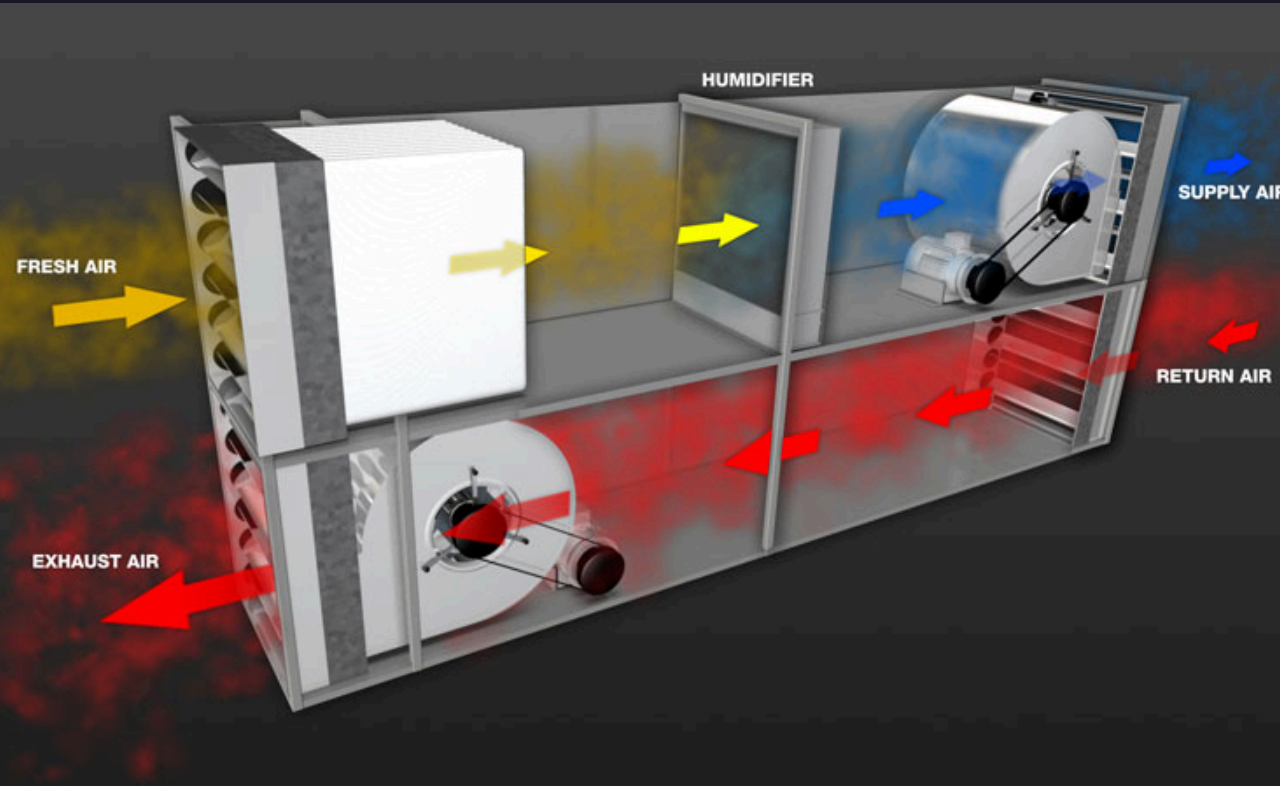
Metrics	Rating
Energy Efficiency ⚡	5/5 
Sustainability 🌿	2/5 
Initial Investment 📷	5/5 

Direct

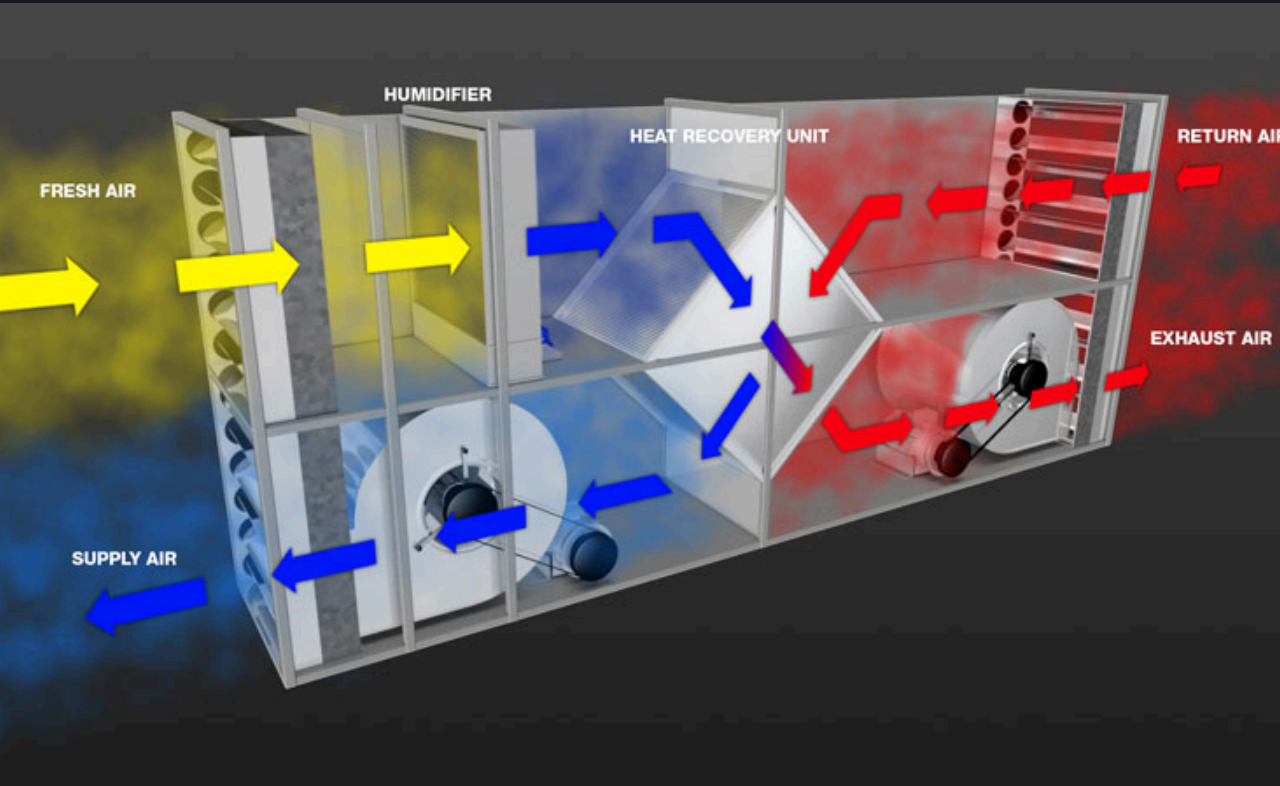
The humidifier adds water to the incoming air stream:

- raising its humidity
- reducing its temperature⁵⁰

This conditioned air is then simply supplied to the room with a high percentage of the room air being exhausted rather than re-circulated to maintain a low humidity in the room.



Indirect



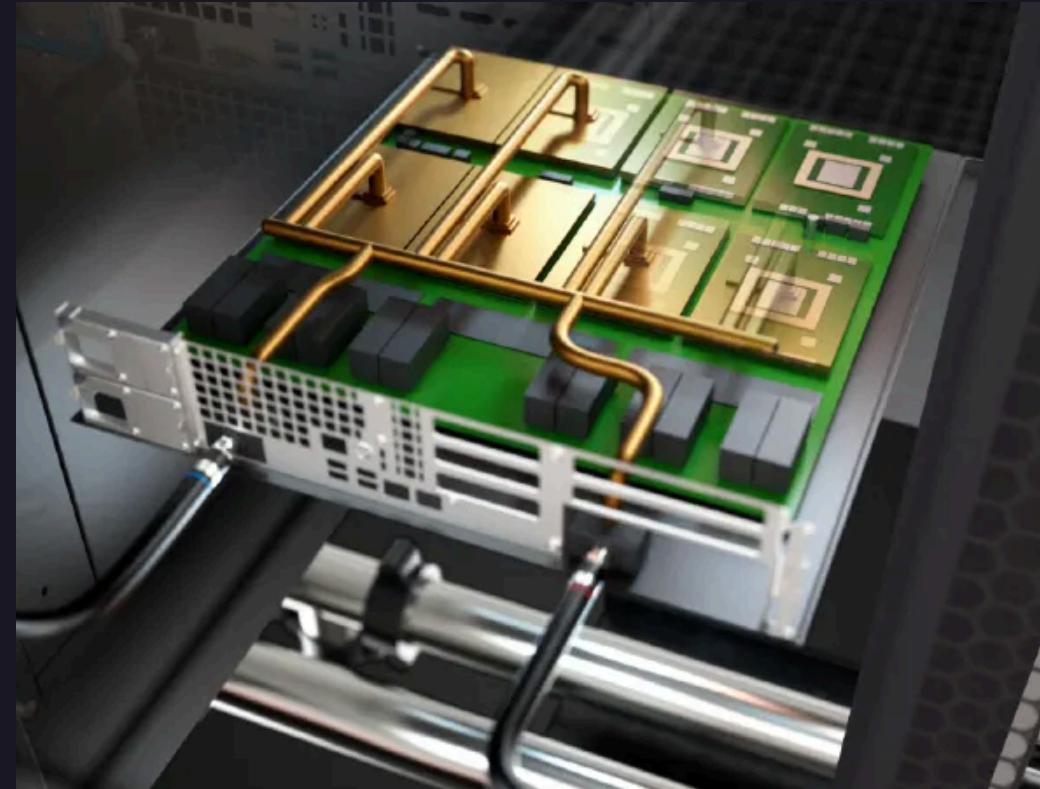
Cool outside air is drawn through a heat recovery unit then immediately exhausted. Meanwhile internal air is drawn from the room and circulated through the heat recovery unit before being re-introduced to the room. The outside air's cool thermal energy is thus transferred to the internal air via the heat recovery unit without the two streams directly mixing

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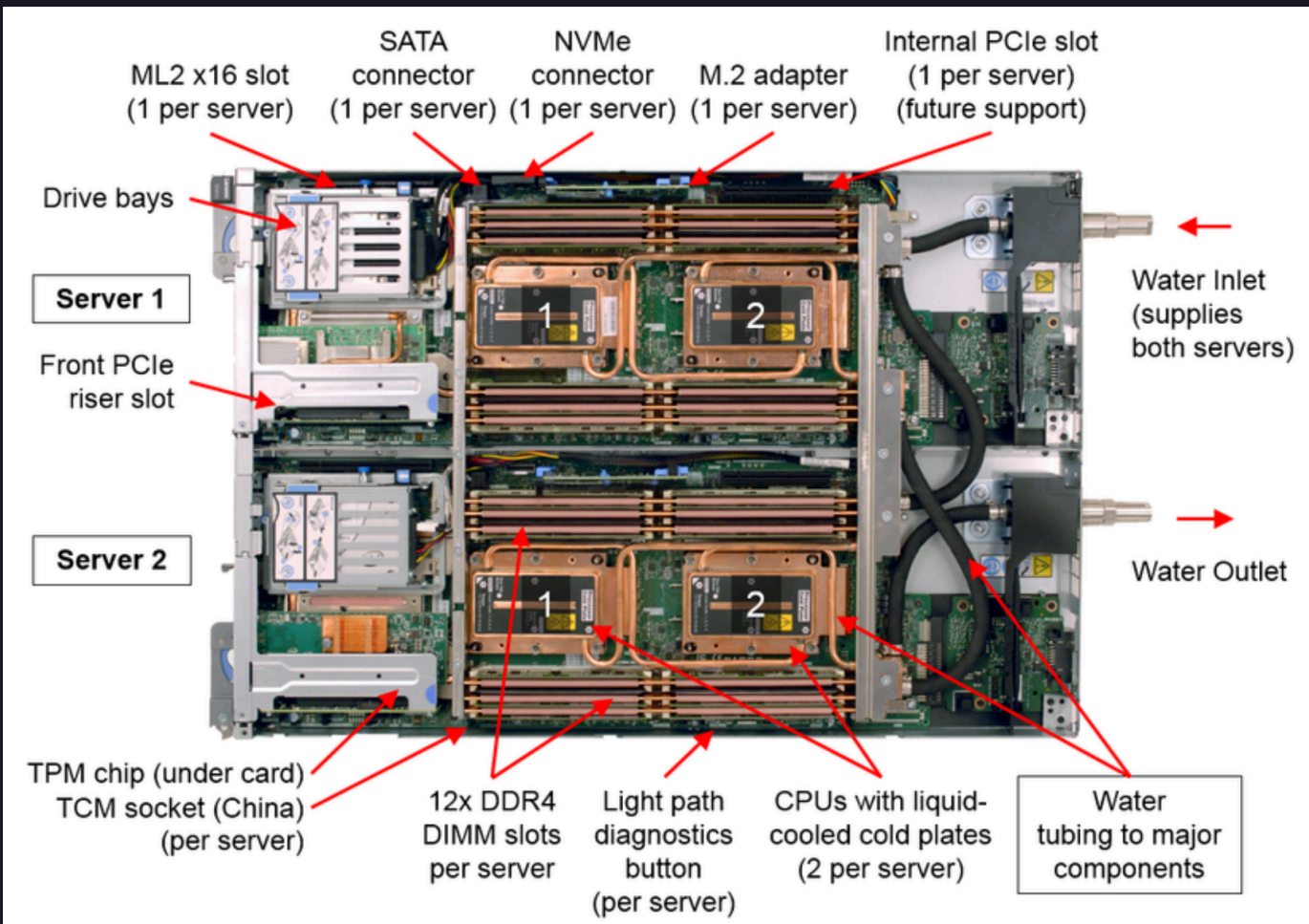
Direct-To-Chip Cooling

D2C systems circulate chilled liquid through pipes near heat-generating components, such as CPUs and GPUs, cooling only areas that require it.

Metrics	Rating
Energy Efficiency ⚡	5/5 ■
Sustainability 🌱	5/5 ■
Initial Investment 📷	2/5 ■



Prone to system disruption (coolant leakage)



Close up Look

1 MW of server equipment costs



approximately 650,000\$

Single-Phase Immersion Cooling

Submerges IT equipment directly in non-conductive liquid that efficiently absorbs and dissipates heat. The leakage risk is lower than D2C cooling.



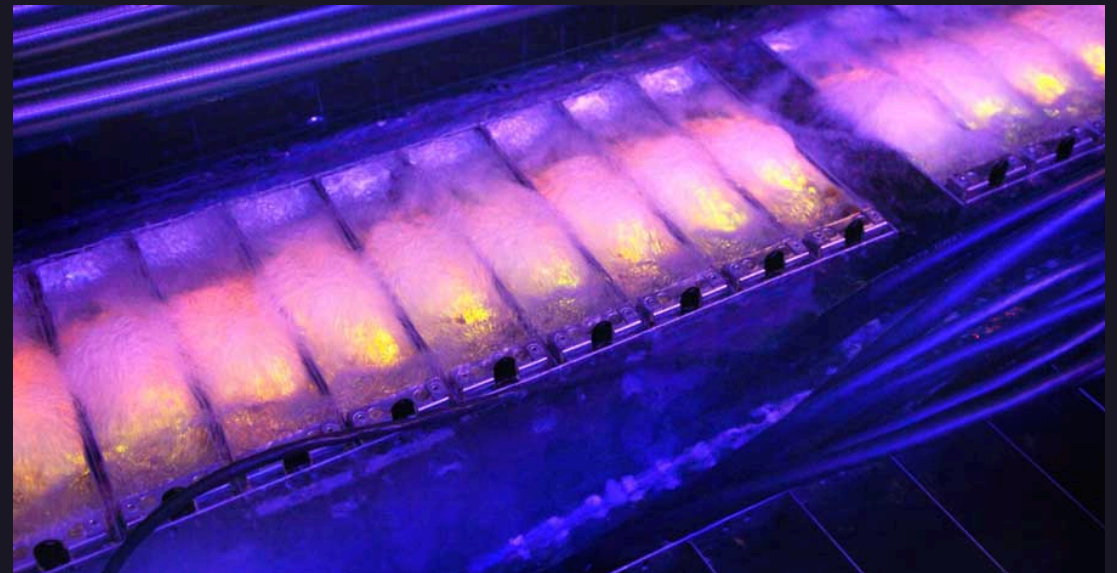
Metrics	Rating
Energy Efficiency ⚡	5/5 ■
Sustainability 🌱	5/5 ■
Initial Investment 📷	1/5 ■

1 MW of server equipment costs approximately 1 million \$

Multi-Phase Immersion Cooling

Improves on the previous method by replacing the liquid with a non-conductive liquid that boils when exposed to the equipment's heat. As the liquid evaporates, it efficiently removes heat, making it one of the fastest and most effective, most expensive cooling solutions available.

Metrics	Rating
Energy Efficiency ⚡	6/5 ■
Sustainability 🌱	5/5 ■
Initial Investment 📷	0/5 ■



Network topology

Network topology optimization

How are the cables routed

Environmental Impact

The disruption observed to the nature's equilibrium

Emissions

Diesel generators and Gas Turbines emit:

- Nitrogen oxides (NO_x)
- Carbon dioxide (CO_2)
- Carbon monoxide (CO)
- Sulfur oxides (SO_x)
- Unburned Hydrocarbons
- Smoke ⁵¹

Causing:

- Health problems
-

Water Shortage 💧

Most servers follow the **Evaporative Cooling** route because it has great energy efficiency while being cost effective. This means:

- Most of the water turns into steam
- Then released into the atmosphere
- And taken out of immediate human usage

Leading to reduced availability for agriculture, drinking, and ecosystems.



Health Effects

The risks datacenters pose to the public's well being

NO_x - Nitrogen Oxides

Methane gas turbines pump harmful nitrogen oxides (mainly NO_2) into the air, which are known to cause:

- Lung irritation
- Breathing difficulties
- Asthma
- Respiratory diseases
- Cancer ^{4, 26, 6}

Children are especially susceptible to respiratory conditions.



Sustained audible noise pollution 🦻

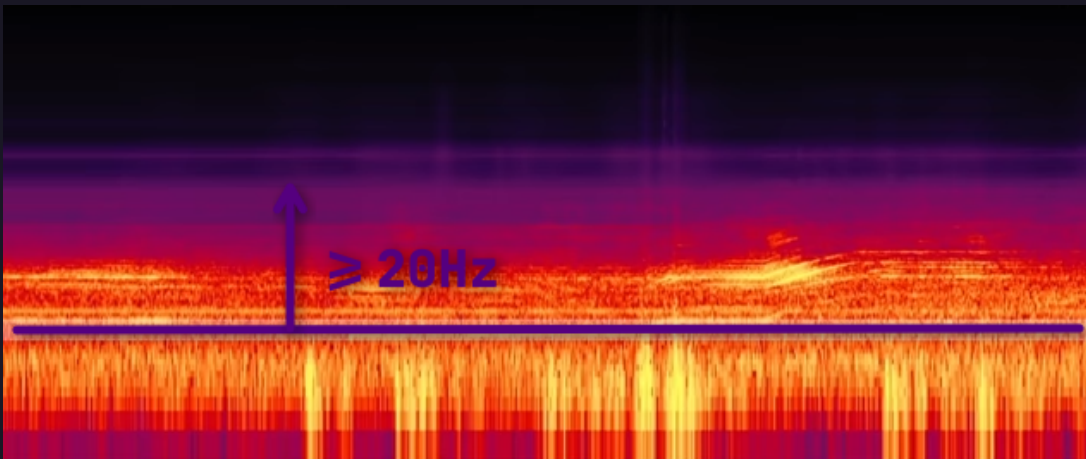
Constant sound exposure is associated with:

- Annoyance
- Disrupted sleep cycles
- Cardiovascular disorders
- Impaired cognitive development of children
- Worsened psychological and physical well-being³⁵
- Seizures (audiogenic epilepsy)³⁶



Infrasound

Infrasound is low frequency sound below the lower limit of human hearing (around 20Hz) ¹



: loud, Y-axis: frequencies, X-axis: time

Known Health Effects ¹³

- Spike in cortisol levels (stress, hypertension) ^{14, 15}
- Nausea, Dizziness ¹²
- Vibroacoustic disease ¹⁶
- High frequency hearing loss ¹⁷
- Shortness of breath ¹⁸
- Anxiety, Depression ¹⁹

Finance

The economic side of the story

Maximized Demand

In the process of building these Datacenters, companies are:

- Placing open-ended orders for memory effectively →
- Telling manufacturers they will buy as much as can be delivered, regardless of the price

Manufacturers are in return:

- Pivoting their factories away from making standard consumer memory to →
- Prioritize high-profit enterprise chips (This affects both **RAM** and **Hard Disks**)

The AI build-out is colliding with a supply chain that cannot meet its physical requirements. ^{46, 47}

Sociopolitical Aspects

The effects that reverberate throughout the

Job Creation

Stargate

- **100,000–200,000** construction and operations jobs ⁹
- **~25,000** onsite jobs ¹¹
- Large automated data centers often need *only dozens* of permanent staff once built
- May only employ **~57** ongoing workers, despite promises of thousands ¹²



Post AI World jobs

Many companies announce layoffs to cut costs tends to show that AI is a revolution like no other, one that doesn't create more jobs than it replaced. The paper with the predictions



AI Occupational Exposure

Tasks which are heavily reliant on the standardized processing of textual information, are things AI is likely to be good at. ⁴⁸

Tasks, which rely on in-person physical dexterity, are naturally less exposed to AI. (recent advancements in robotics question this)

Table 1: Occupations with the highest and lowest AIOE measures

Rank	Highest scoring	Lowest scoring
1	Genetic counselors	Dancers
2	Financial examiners	Fitness trainers
3	Actuaries	Helpers - painters, paperhangers, plasterers, and stucco masons
4	Purchasing agents, except wholesale, retail, and farm products	Reinforcing iron and rebar workers
5	Budget analysts	Pressers, textile, garment, and related materials

Source: Table 2 from Felten et al. (2021)

Advancements in Robotics

China CGTN 2026

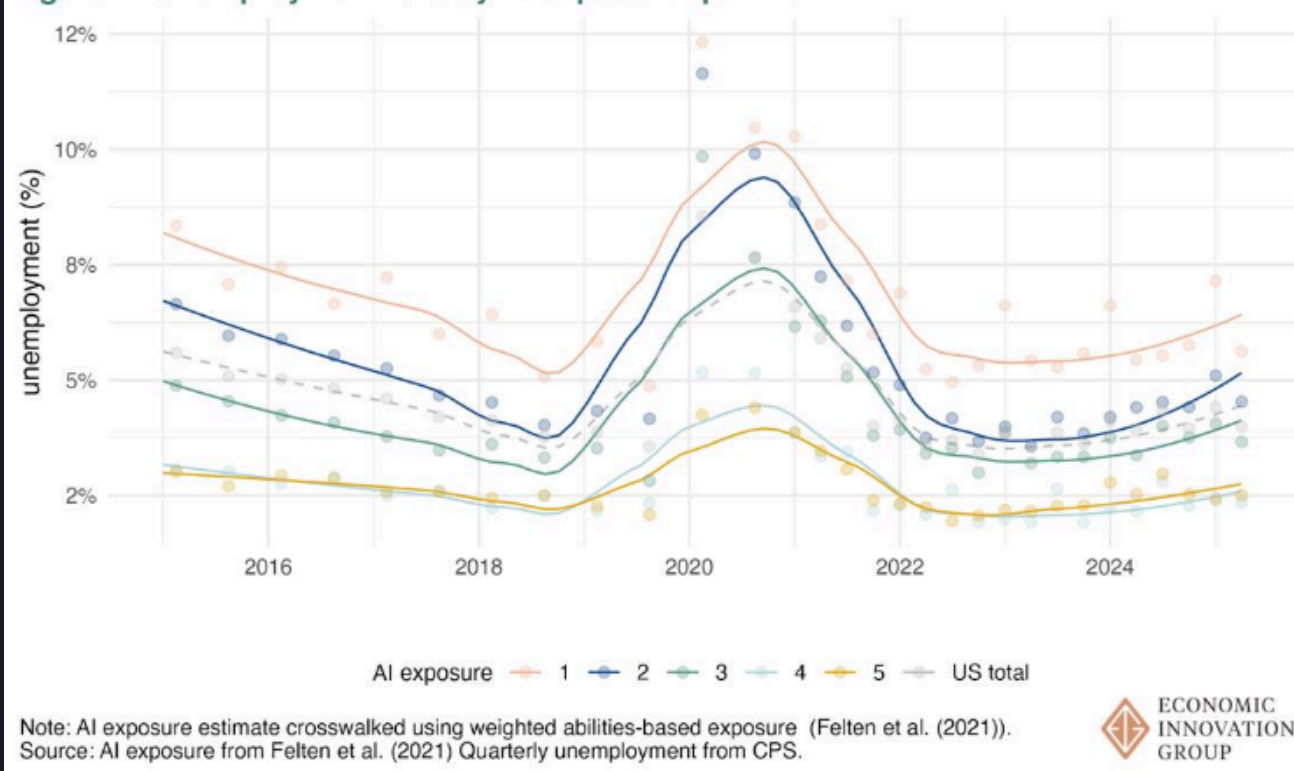
On China's Spring festival (Chinese New Year) we witnessed some impeccable AI augmented robot dexterity.

Growth is rapid and real life applications are imminent.

▶ 0:00 / 4:50



Figure 1. Unemployment rate by AI exposure quintile



Unemployment

5 equally sized groups based on their AIOE score, with quintile 1 being the least AI-exposed and quintile 5 being the most AI-exposed

More unemployment for the least exposure workers. ⁴⁸

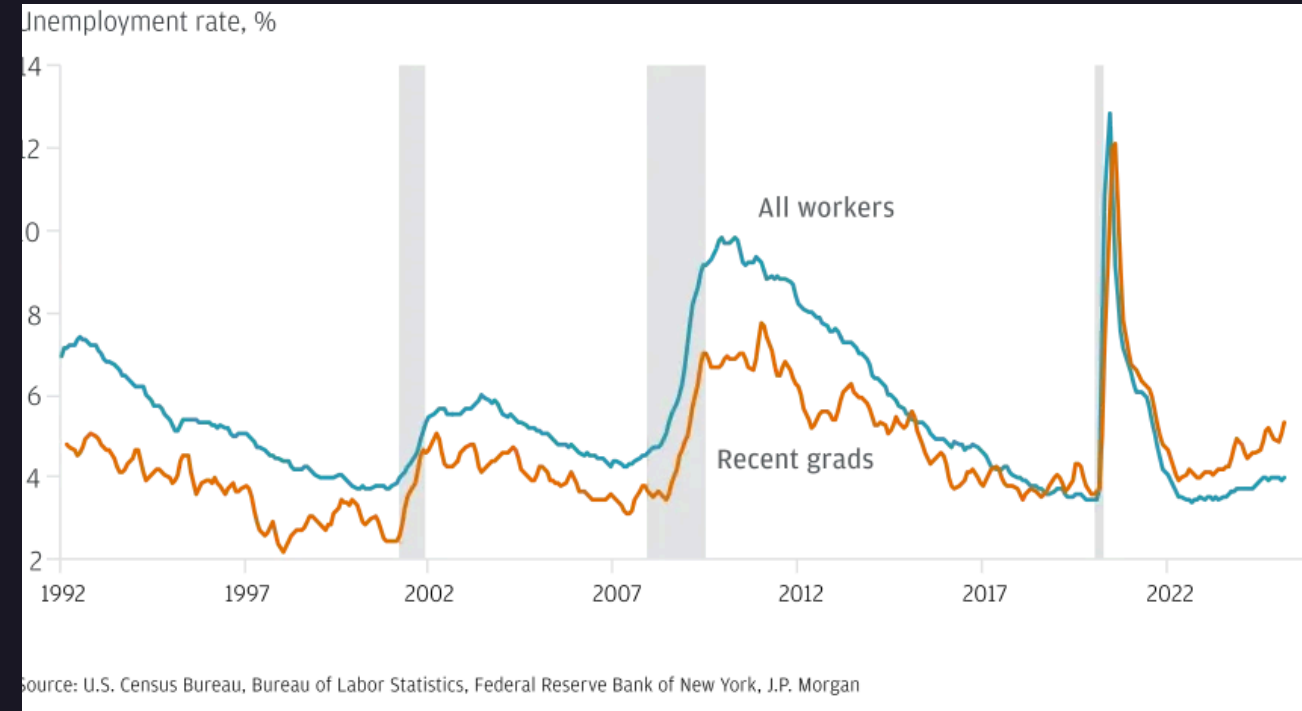
Opposite point of view

Compared to previous technological revolutions, experts say AI is chipping away at many white-collar jobs, including coding and marketing roles, and allowing companies to reduce hiring and improve productivity with fewer workers.⁵⁴



J.P. Morgan study

- Graduate students face a tougher job market
- In affected professions experience is appreciated ⁵⁵



Backlash



Across the nation many revolts against datacenters are occurring.

More than 50 active groups across 17 states targeted 30 projects during that time period, two-thirds of which were halted.⁷

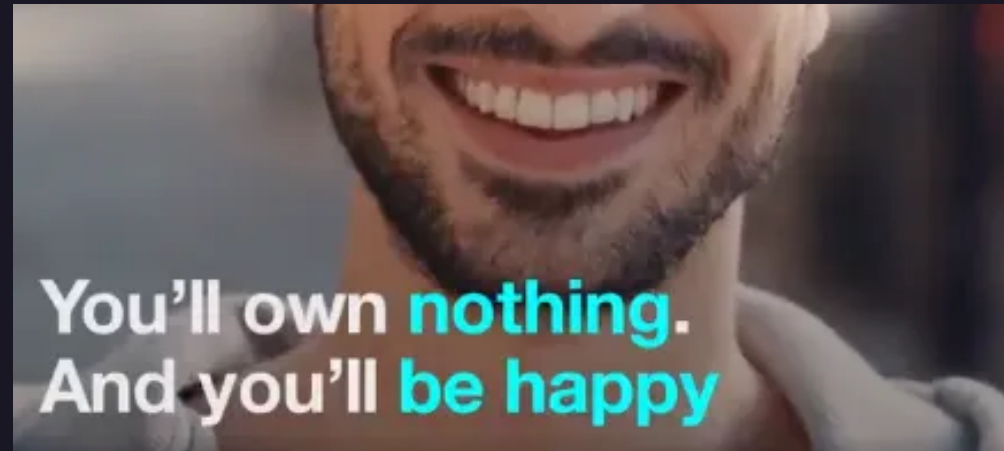
“Monterey park protest”

Ownership

Shift towards cloud computing

- Less burden on the user's device 
- Can sell less powerful devices, useless without cloud computing assistance ⁴⁰
- Cloud storage, cloud gaming ⁴³, cloud movies ⁴¹, cloud music ⁴², cloud AI, cloud everything 

Cloud services instigate subscription based models which degrade ownership.



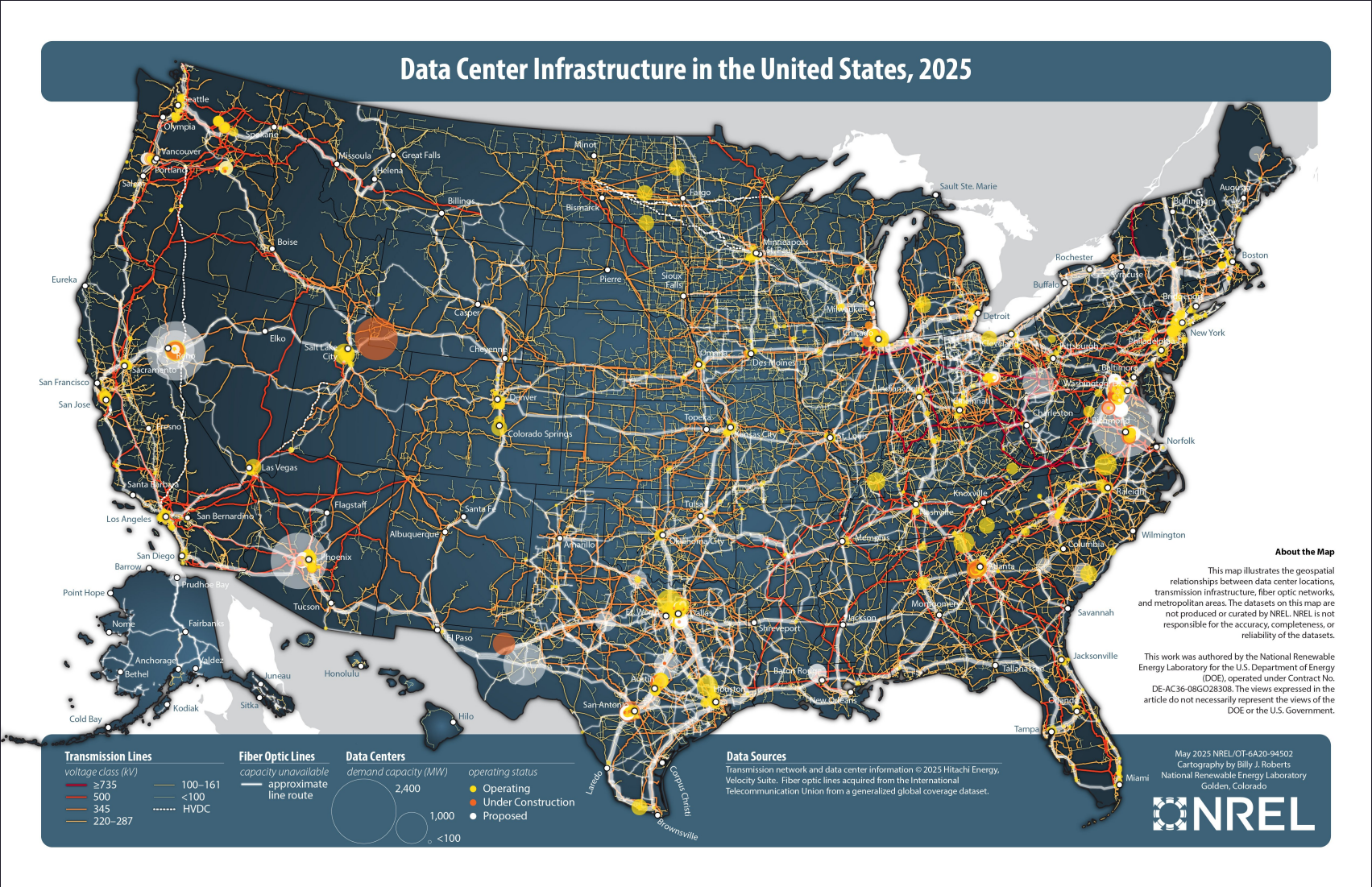
Datacenter Case Study

The datacenters that are responsible for running the world's cloud services and facilitating the training of the most advanced LLMs

Map

Most Activity

	State
1	Virginia
2	Texas
3	California
4	Illinois
5	Ohio



LLM Facilities 🏭

Datacenter	Location	Company	LLM
Stargate Project	Michigan	OpenAI, Oracle	ChatGPT
Colossus	Memphis	xAI	Grok
Hyperion	Louisiana	Meta	Llama
Google Datacenter	Iowa	Google	Gemini
Fire-Flyer AI Cluster	Zhejiang	Fire-Flyer	Deepseek





Colossus

Specifications

Specs	Value
Investment	
Location	Memphis, Texas
Company	xAI
IT Capacity	

Good example because:

- AI usage
- centralized to one datacenter

Server Hardware

**~110,000 NVIDIA GB200 NVL72 GPUs (targeting 1.1 PFlops FP8 compute).
usually small fp numbers needed for**

Methane Gas Turbines 🛢️

Specifications:

- natural gas turbines for primary generation
- batteries for stability
- grid for long-term scalability

Elon Musk's xAI data center Methane gas turbines, Colossus. ²⁴

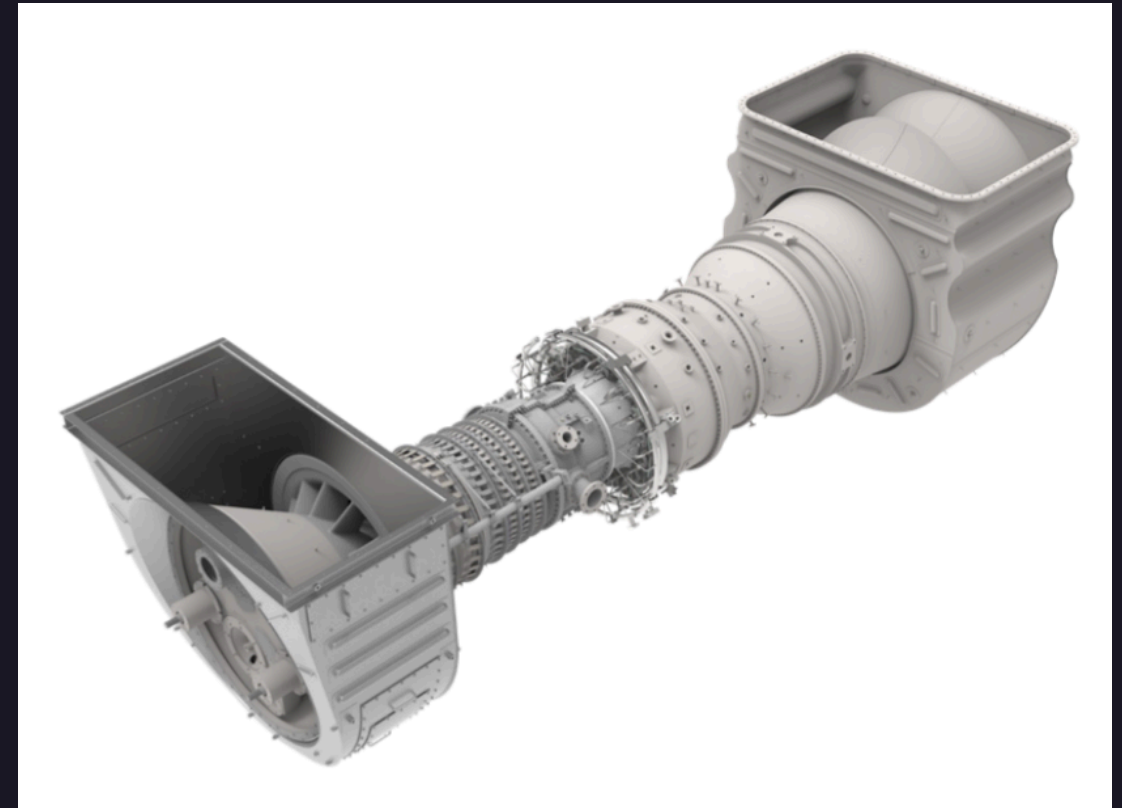


- 7 Solar Turbines Titan-350 ²²
- 12 Solar Turbines SMT-130 ²³

Solar Turbines: a Caterpillar subsidiary

Solar Turbines Titan 350

Specifications	Value
Power output	35-39MW
Thermal efficiency	~40%
Fuel types	Natural gas, Propane, Low BTU gases



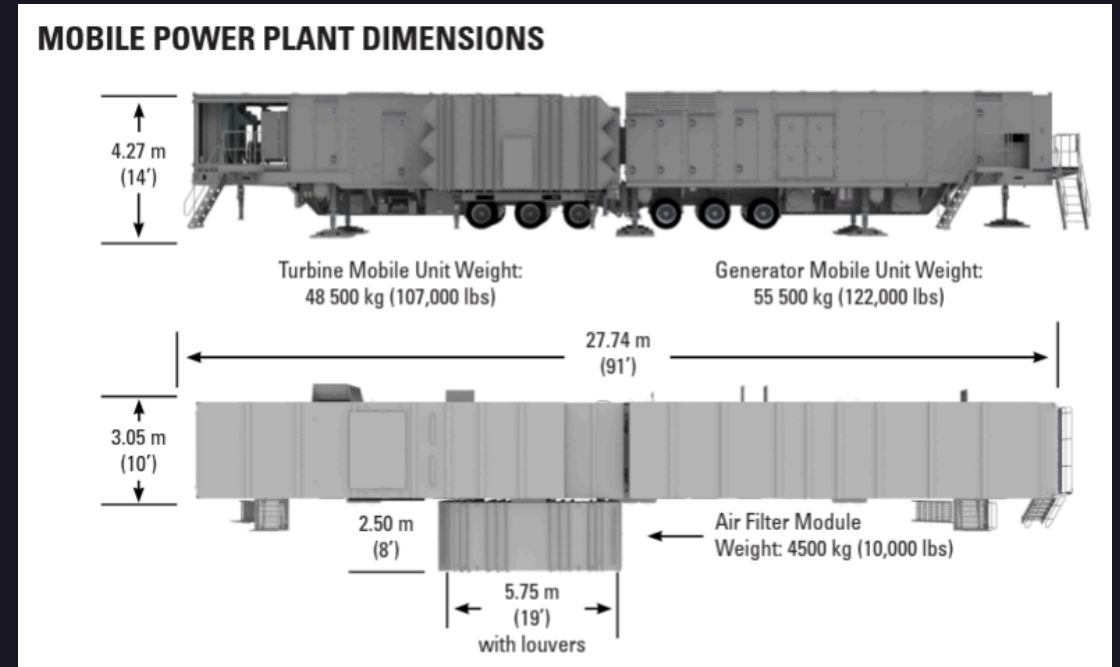
source^{22, 24, 27}

Solar Turbines Titan 350

Advanced Specifications	Value
Heat Rate	8845 kJ/kW-hr - 8780 kJ/kW-hr
Exhaust Flow	371980 kg/hr - 387820 kg/hr
Exhaust Temp	460°C - 490°C
Emissions	25 PPM NOx

Solar Turbines SMT 130

Specification	Value
Power output	16MW
Thermal eff.	~36%
Fuel types	Field Gas, CNG, LPG
Heat Rate	10160 kJ/kWe-hr
Exhaust Flow	202510 kg/hr
Exhaust T	490°C
Emissions	25 PPM NOx



Fully-Integrated Mobile Power Plant
Powered by Titan 130 ^{23, 24, 28}

+ Added Diesel fuel support

SoloNOx

SoloNOx is the technology enabling Solar Turbines to reduce NOx and CO emissions.³⁰

Offers a robust:

- 9ppm NOx,
- 15ppm CO, and
- 15 ppm UHC

emissions warranty for natural gas fuel.

Available power

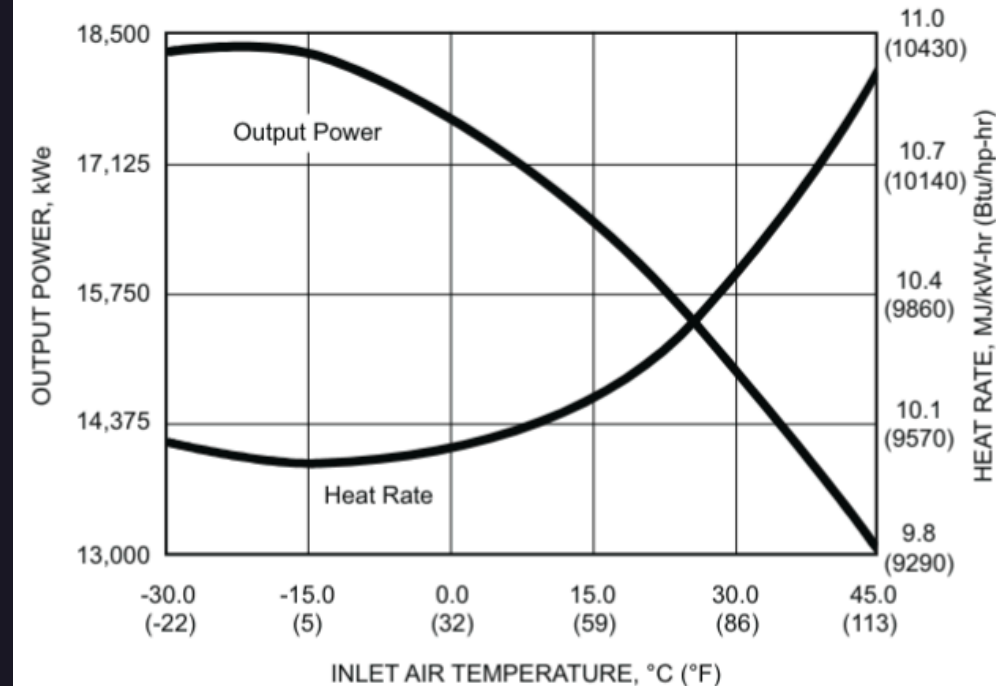
Titan 130 available power graph

As inlet air temperature increases the power the generator can produce decreases.

Why?

Colder inlet air means higher mass flow rate which in turn increases the power output ⁵²

Available Power*



* SoLoNOx

Tesla Megapacks

Megapacks ²⁹ are large scale energy storage. *Many many Batteries*

Used for:

- Power Backup
- Resilience for outages
- Demand-response
- Turbine ramping

They can provide 4 hours of power. They are planing to deploy 200 Megapack for about 1 gigawatt hour of buffering. Long term solar powered facility but near-term it is Megapacks and gas turbines that is being used as a bridge to the future.

Sustainable Pledges

xAI plans an 88-acre solar array to provide power to Colossus ⁴⁴

Legally Ambiguous Operation

Electricity-generating turbines were exempt from requirements for air quality permits.

Loophole: Generators can operate without permits as long as the machines don't remain stationed for ≥ 1 year.

xAI eventually received permits for 15 turbines at Colossus 1 and is now operating 12 permitted machines at the site.

Net annual nitrogen oxide emission reductions of up to 296 tons by 2032," ⁴

Ambitious Projects

Colossus

Colossus power generation around 1.1 GW by Q2 2027 and 2 GW total capacity

Hyperion

- ~2 GW by 2030 and end goal 5GW. While Meta has agreements for 1.8 GW of solar and wind, 150 MW geothermal, and even nuclear power extensions, most current power is set to come from gas ⁴⁵



Stargate

Specifications

Specs	Value
Investment	\$500 billion
Location	Abilene, Texas
Company	OpenAI & Oracle
IT Capacity	4.5 GW

based on ¹⁰

5 new sites

- Shackelford County, Texas
- Milam County, Texas
- Doña Ana County, New Mexico
- Lordstown, Ohio
- a mystery site located somewhere in America's Midwest.

End goal being 10GW of combined capacity.

Chips

- Nvidia GB200 racks